

PRADYUMNA UPADHYAY

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EDUCATION

University of Maryland, College Park, MD | *Master of Engineering in Software Engineering*
Devi Ahilya Vishwavidyalaya, Indore, India | *Bachelor of Computer Engineering*

Aug 2023 - May 2025
Jul 2017 - Jun 2021

SKILLS

- **Languages & Fundamentals:** Python, C++, C#, JavaScript, TypeScript, Data Structures, Algorithms
- **Web Development & APIs:** React, Redux, Angular, Node.js, Flask, FastAPI, HTML/CSS, jQuery, OAuth, REST APIs
- **Cloud & DevOps:** AWS (EC2, EMR, S3, Cloud9), Git, GitHub, CI/CD, Postman, VS Code
- **Databases & ORM:** MySQL, MSSQL, MongoDB, SQLAlchemy
- **Big Data & Geospatial:** PySpark, Pandas, GeoPandas, ArcGIS, Folium, Geopy, OpenStreetMap, Esri Builder
- **AI & Systems:** LLMs, Prompt Engineering, NLP, Async Programming, Retry Handling, Caching, Modular Design, Fault Tolerance
- **Architecture Concepts:** Client-Server, Batch Processing, Distributed Computing, Session Management, Horizontal Scaling
- **Tools & Workflow:** Agile, JIRA, Confluence, Slack

WORK EXPERIENCE

University of Maryland | **Software Engineer Intern**

Oct 2023 - Present

- Built **full-stack geospatial web applications** using **Flask, React**, and **AWS**, leveraging a **client-server architecture** to enhance accessibility of transportation safety data across Maryland.
- Developed **RESTful APIs** and **session management** to streamline crash data queries between backend and frontend.
- Optimized **PySpark-based batch jobs** on **AWS EMR** for pedestrian datasets, leveraging **distributed data processing** to reduce execution time by **30%** for over **20TB** of geospatial data.
- **Reduced data interpretation errors by 25%** through an interactive Folium/OpenStreetMap visualization for high-risk pedestrian areas.
- Created **modular backend services** with **real-time export functionality**, enabling dynamic **PDF and Excel report generation**.
- Leveraged **caching mechanisms** to minimize redundant inputs and improve **user experience by 50%**.

LTIMindtree | **Software Engineer**

Aug 2021 - Jul 2023

- Maintained **production system** across **7 countries** for **200+ users**, ensuring **stable operations** and prompt incident resolution.
- **Refactored SQL Server database architecture**, reducing **data upload latency by 85%** and shrinking **database size by 15%** in an ASP.NET MVC-based pricing application.
- Improved **cross-team efficiency by 30%** by bridging **Dev, QA, and DevOps** teams, ensuring rapid issue resolution.
- Decreased **bug frequency by 25%** and accelerated **release cycles by 30%** via Agile-driven SDLC improvements.
- Mentored and trained **4 junior engineers** through documentation, walkthroughs, and **onboarding sessions**, accelerating their ramp-up time.

PROJECTS

Travel Buddy

Bitcamp 2025

- Built a **full-stack AI-driven travel assistant** using **FastAPI and React**, in under **36 hours**, delivering real-time personalized itineraries and contextual travel insights.
- Integrated **third-party APIs** using **fault-tolerant API orchestration** with **max-retry logic** to ensure reliable data retrieval.
- Designed and deployed a **responsive UI** with **Tailwind CSS** and **loaders**, enhancing usability during high-demand use cases.

UpSkilled - Learning Management System

Aug 2024 - Dec 2024

- Developed a **microservices-based LMS platform** using **React and Spring Boot** with dashboards tailored for **instructors, admins, and employees**.
- Embedded **assignment viewers** and streamlined **grading workflows**, improving **instructor efficiency by 40%**.
- Increased **course completion rates by 25%** by enabling **intuitive tracking of progress**, and access to enrolled content.

SmartCam - Facial Recognition Attendance System

Aug 2024 - Dec 2024

- Co-created a **dual application system: Python/OpenCV-based desktop attendance app** with **95% face recognition accuracy** under varying light conditions.
- Delivered a **web dashboard** with **React/Node.js** enabling admins and faculty to **manage, and analyze attendance metrics**.
- Integrated **MongoDB Atlas** for **real-time log storage** and scalable reporting.